

A Parameter-Free Radial Acceleration Relation from the Dark-Energy Scale

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Abstract

We remove the MOND acceleration scale as an independent parameter. If the interpolating function’s argument is the acceleration measured in units of the dark-energy acceleration $s = c\sqrt{G\rho_\Lambda}$ rather than in units of a_0 , then a_0 is an output and the coefficient κ in $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ is exactly the reciprocal of the function’s deep-MOND slope. Every parameter-free interpolating function in common use has slope one, predicting $\kappa = 1$, which two independent measurements exclude at $7\text{--}10\sigma$. We exhibit a family, $\mu_n(Y) = 1 - (1 + Y)^{-n}$, whose deep-MOND slope is its integer exponent, so the slope is a count rather than a dial; it obeys the composition law $1 - \mu_n = (1 - \mu_1)^n$ and is, to our knowledge, new. Each integer is then a complete prediction of the radial acceleration relation with *nothing* fitted — not the scale, not a shape parameter. On 155 SPARC rotation curves (2788 points, standard mass-to-light ratios) the data select $n = 2$: rms residual 0.1502 dex against 0.1660 for $n = 1$ and 0.1853 for $n = 3$ on the dark-energy convention, and 0.1438 against 0.1846 and 0.1684 on the critical-density one. For comparison ν_{RAR} , *fitted* at a *fitted* scale, gives 0.1453 dex on the same data. The two acceleration footings this programme carries are exactly $n = 2$ on the two density conventions, to 0.02% and 0.33%. The curve predicts a *power-law* rather than exponential approach to Newton, leaving a solar-system anomaly of $5.8 \times 10^{-16} \text{ m s}^{-2}$ at Saturn ($17\times$ below the Cassini residual, where the fitted kernel predicts zero); a wide-binary boost differing from the fitted kernel by up to 0.28 in γ_v ; a Tully–Fisher zero point with no freedom; and an a_0 that inherits the dark energy’s equation of state, flat for $w = -1$ and $1.31\times$ higher at $z = 5$ for $w = -0.9$. **We do not derive the integer.** Four structural angles were searched and none fixes it; two point elsewhere (the algebraically clean member is $n = 3$; degree-of-freedom counting wants $3/2$; the AQUAL–QUMOND duality favours $n = 1$). The data select the exponent and the mathematics does not, and we report that as prominently as the fit.

1 The scale as an output

MOND introduces one constant, $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$, numerically close to $c\sqrt{G\rho_\Lambda}$. That closeness is usually recorded as a coincidence [2]. We take it as a definition instead. Write the AQUAL field equation [1] with the interpolating function’s argument in units of

$$s \equiv c\sqrt{G\rho_\Lambda} = 1.873 \times 10^{-10} \text{ m s}^{-2}, \quad (1)$$

so that the theory contains no independent acceleration scale:

$$\nabla \cdot [\mu(|\nabla\Phi|/s) \nabla\Phi] = 4\pi G\rho. \quad (2)$$

In the deep limit $\mu \rightarrow cY$ with $Y = |\nabla\Phi|/s$, the spherical solution of (2) gives $|\nabla\Phi| = \sqrt{GMs/c}/r$. Matching to $\sqrt{GMa_0}/r$,

$$\boxed{a_0 = s/c, \quad \kappa \equiv a_0/\sqrt{G\rho_\Lambda} = 1/c} \quad (3)$$

The coefficient is the reciprocal of the function’s deep-MOND slope, and nothing else. Every parameter-free interpolating function in use — $Y/(1+Y)$, $Y/\sqrt{1+Y^2}$, $1 - e^{-Y}$, $\tanh Y$ — has slope exactly one, because a function with no free parameter turns over where its argument is one. All four therefore predict $\kappa = 1$, which this programme’s two independent measurements, 0.465 ± 0.076 and 0.551 ± 0.043 , exclude at 7 and 10σ .

2 A family whose slope is a count

$$\mu_n(Y) = 1 - (1 + Y)^{-n} \quad (4)$$

has $\mu'_n(0) = n$ exactly and $\mu_n(\infty) = 1$ for every n . The slope is therefore not a dial but an integer, and the family obeys

$$1 - \mu_n = (1 - \mu_1)^n, \quad (5)$$

so n is a composition count: “at least one of n identical channels responds”, with $\mu_1 = Y/(1+Y)$ the single-channel response. To our knowledge (4) is not a previously proposed interpolating function; its $n = 1$ member is the known “simple” μ [3] and its $n \rightarrow \infty$ limit the “exponential” one, but the family is distinct from the standard $\mu_n = y/(1 + y^n)^{1/n}$, coinciding with it only at $n = 1$ (the first correction appears at different order, so it is not a reparametrisation).

3 Every integer is a complete prediction

With (4) in (2) and s fixed by the cosmology, an integer n predicts the entire radial acceleration relation with no fitted quantity. We test on the SPARC rotation curves [4] with the standard $\Upsilon_{\text{disk}} = 0.5$, $\Upsilon_{\text{bulge}} = 0.7$, keeping curves with ≥ 3 points at $\delta V/V < 0.10$: 155 of 175 curves, 2788 points.

n	$\kappa = 1/n$	a_0 predicted	rms (ρ_Λ)	rms (ρ_{crit})
1	1.000	1.8725×10^{-10}	0.1660	0.1846
2	0.500	9.3623×10^{-11}	0.1502	0.1438
3	0.333	6.2415×10^{-11}	0.1853	0.1684
4	0.250	4.6812×10^{-11}	0.2183	0.1981

The data select $n = 2$ on both conventions. The neighbours are worse by 0.016 and 0.035 dex on the dark-energy convention, and by 0.041 and 0.025 dex on the critical-density one, over 2788 points. Profiling the scale freely at $n = 2$ gives $1.1s$ and improves the scatter only marginally, so the parameter-free version is not carried by luck. For comparison ν_{RAR} , *fitted*, at a *fitted* scale, gives 0.1453 dex on the same data. The parameter-free prediction is within 0.005 dex of it on the dark-energy convention and *below* it, 0.1438, on the critical-density one — a fit-free curve marginally outperforming a fitted one, which we note without weight, since 0.0015 dex is far inside the systematics of fixed mass-to-light ratios.

A consistency check worth stating: this programme’s two registered acceleration footings, 9.3619×10^{-11} and 1.1279×10^{-10} , are exactly $s/2$ on the dark-energy and critical-density conventions, to 0.02% and 0.33%.

4 Predictions

A power-law approach to Newton. $1 - \mu_2 = (s/g)^2$, where ν_{RAR} leaves $e^{-\sqrt{g/a_0}}$. At Saturn’s orbit that is a fractional departure of 9.7×10^{-12} , an anomalous acceleration $5.8 \times 10^{-16} \text{ m s}^{-2}$, about $17\times$ below the Cassini residual. The fitted kernel predicts *zero* there; this one does not.

Wide binaries. $\gamma_v(20 \text{ kAU}) = 1.750$ against 1.928 for ν_{RAR} , with differences up to 0.281 across 2–30 kAU — far above what the astrometric sample can resolve. (This programme’s frozen wide-binary preregistration is untouched by this work.)

Tully–Fisher. $V^4 = GMa_0$ with a_0 predicted: zero-point offsets of -0.027 and -0.006 dex relative to the fitted scale. The normalisation stops being a calibration.

The acceleration scale inherits w . $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$: exactly flat for $w = -1$, and $1.31\times$ higher at $z = 5$ for $w = -0.9$. A measurement of a_0 at high redshift is a measurement of the dark energy’s equation of state.

5 What we do not have

We do not derive the integer. Four structural angles were searched and none fixes it. *Resummation:* the composition law is exact and n is a channel count, but no construction fixes the count, and the real-versus-complex field convention already slides it by two. *Dimension counting:* spatial dimension fixes only the deep-limit power (the p -Laplacian at $p = D$), which is n -blind since every member is linear at small argument. *Phase space:* the family is exactly the exponential kernel smeared over a Gamma-distributed rate, with n a dispersion; counting degrees of freedom gives $n = d/2 = 3/2$ in three dimensions, *not* two. *Duality:* no interpolating function is self-dual, and only $n = 1$ has a closed-form AQUAL–QUMOND dual.

Worse for the reading we would like: integrating (4) shows the *algebraically* distinguished member is $n = 3$, the only one of the first three with a rational free function, while $n = 2$ shares a logarithm with $n = 1$. The one fact specific to $n = 2$ is that it is the *marginal* index — $\int(1 - \mu_n)$ converges only above it — which is the kind of boundary a criticality argument might select, but a boundary case is not a derivation.

So: one empirical argument for $n = 2$, from rotation curves with nothing fitted, and three structural arguments pointing at other integers. **The data select the exponent; the mathematics does not.** No published work derives an interpolating exponent at all [5].

6 What is not claimed

This is single-field AQUAL and carries none of the relativistic sector. The SPARC comparison uses fixed mass-to-light ratios with no per-galaxy nuisance parameters and is an rms comparison, **not a likelihood**; a full analysis marginalising distance, inclination and mass-to-light would move the numbers, and points are pooled rather than weighted per galaxy. Four integers were tested.

The wide-binary estimate uses the simplest additive external field with no angular structure. The Cassini comparison uses a representative residual bound, not a published covariance. The predicted a_0 sits 6–28% *below* the literature’s fitted value on the two conventions; that is not a contradiction, since a fitted a_0 is kernel-dependent, but it is not agreement either.

7 Reproducibility

`fable_independent_2026/` lanes L226–L234 with committed outputs and results files. The SPARC test is `L232_sparc_parameter_free.py`; predictions `L233_predictions.py`; the negative structural search `L234_structural_search.py`. Lean file `lean_2026/Mondlean.lean`: 158 theorems, exit 0, zero `sorry`.

References

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- [4] F. Lelli, S. McGaugh, J. Schombert, *Astron. J.* **152**, 157 (2016).
- [5] Modified Newtonian Dynamics review (2025), [arXiv:2501.17006](https://arxiv.org/abs/2501.17006).
- [6] The research repository, <https://github.com/carlzimmerman/zimmerman-formula> (2026).